

CG Height Test Plan

Method A: Side Tilt (Balance) Test

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1 Purpose

This document outlines how we find the center of gravity (cg) height of ARG26 through a physical test. Although we can make reasonable estimates through CAD, validating the cg height is important because it influences a lot of our load transfer and handling characteristics.

2 Method A: Side Tilt (Balance) Test

2.1 Theory

A body balances on a pivot when its CG is directly above the pivot. In this test the car is tipped sideways onto the tires of one side, so it rests only on the outer edges of those two tires. The car is slowly rotated until it balances and can be held in place with a couple of fingers, with the crane chain slack. At that point the CG lies on the vertical line through the pivot.

In the car's own frame, the CG sits half the car's width, $d/2$, in from the pivot edge and a height h above the ground plane. When the car is rotated by the balance angle θ , the line from the pivot to the CG is vertical, so $\tan \theta = (d/2) / h$.

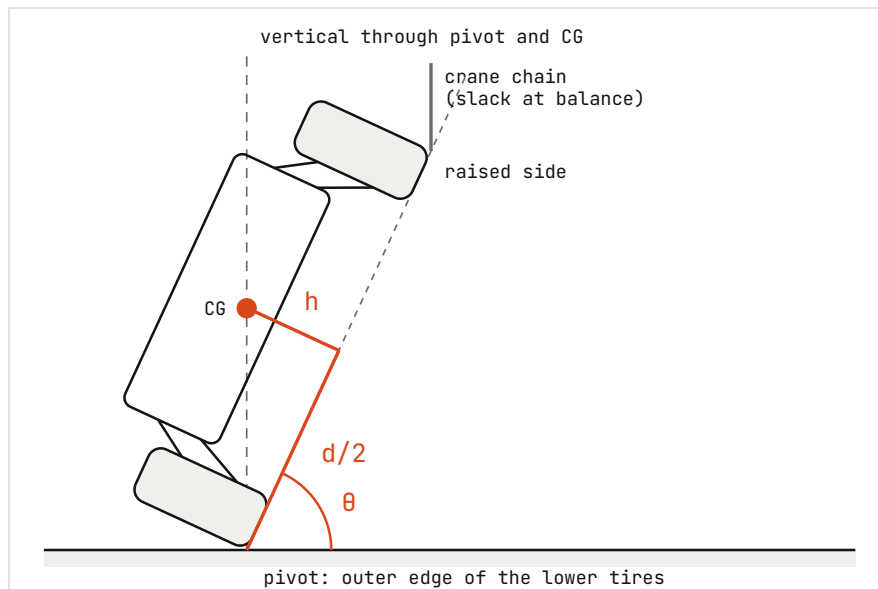


FIGURE 1 Front view of the car balanced on its left tires. At balance the CG is directly above the pivot, so $\tan \theta = (d/2) / h$.

2.2 Materials

#	Item	Where
1	ARG26	Lab
2	Drivers (driver 1 and driver 2)	—
3	Vehicle scales (corner weights)	Lab
4	Digital level	Lab
5	Crane with hook	Lab
6	Chain long enough to wrap around a front rim and a rear rim together, with a maillon on both ends	Lab
7	Dummy shocks and dummy cones	Lab
8	Magnesium rims	Lab
9	Tire pump and pressure gauge	Lab
10	Camera or phone (to video the digital level)	—

2.3 Pre-Test Procedure

1. Install the accumulator and bolt it down (ask someone on high voltage).
2. Install the dummy shocks and dummy cones.
3. Make sure the car is on the magnesium rims.
4. Overinflate the tires to 30 psi.
5. Once the car is on the ground, measure the ride height. It will be larger than usual because the dummy shocks are in.
6. Put the digital level on a flat, visible part of the car and zero it.

2.4 Measurements Before Tilting

Record these with the car level on the scales, once for each test case.

Symbol	Measurement	Without driver	Driver 1	Driver 2
W_{FL}	Front left corner weight (lb)			
W_{FR}	Front right corner weight (lb)			
W_{RL}	Rear left corner weight (lb)			
W_{RR}	Rear right corner weight (lb)			
W_{driver}	Driver weight (lb)	—		
W	Total weight (lb)			

These are the same for all three cases:

Symbol	Measurement	Value
L	Wheelbase (in)	
d_f	Front width, outside tire edge to outside tire edge (in)	
d_r	Rear width, outside tire edge to outside tire edge (in)	
p	Tire pressure (psi)	
RH	Ride height with dummy shocks in (in)	

2.5 Equations

The front and rear of the car have different widths, so the width is taken at the lengthwise position of the CG (Figure 2) [2]. The CG sits a distance a behind the front axle, set by the fraction of weight on the rear axle, $W_R = W_{RL} + W_{RR}$:

$$a = L \times W_R / W$$

$$d = d_f + (d_r - d_f) \times W_R / W$$

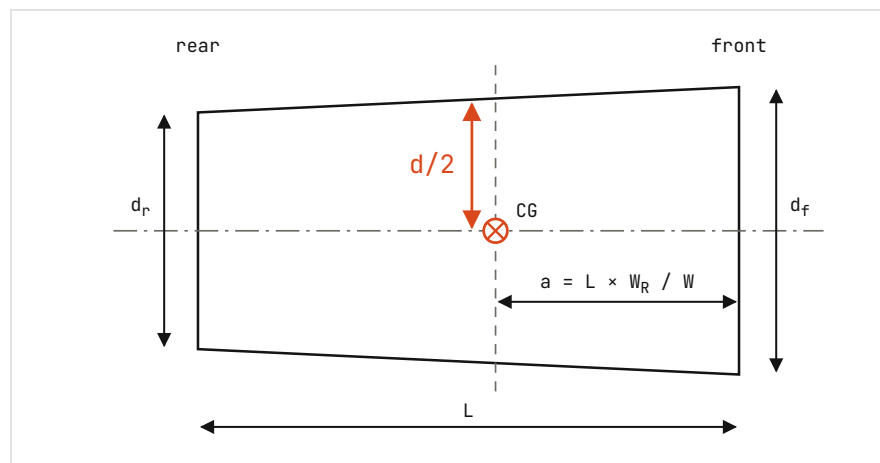


FIGURE 2 Plan view. The width d is taken across the outside tire edges at the lengthwise position of the CG. Adapted from [2].

CG height above the ground at static ride height:

$$h = (d / 2) / \tan \theta$$

where θ is the digital level reading at balance (the level is zeroed with the car flat on the ground). This assumes the CG is on the car's centerline.

Before lifting, estimate the angle the car should balance at from the expected CG height (for example, from CAD), so you know roughly where to stop the crane:

$$\theta_{\text{expected}} = \arctan((d / 2) / h_{\text{expected}})$$

Ride Height Correction

The dummy shocks hold the car above its normal ride height, so the measured h is the CG at the tested ride height. Only the sprung mass moves with ride height, so the CG at the design ride height is:

$$h_{\text{nominal}} = h_{\text{measured}} - (m_{\text{sprung}} / m_{\text{total}}) \times (RH_{\text{tested}} - RH_{\text{nominal}})$$

The driver is part of the sprung mass, so the sprung fraction is different for each case. With W_{unsprung} the same in every case:

$$W = W_{\text{car}} + W_{\text{driver}}, \quad m_{\text{sprung}} / m_{\text{total}} = (W - W_{\text{unsprung}}) / W$$

DERIVATION

Split the car into its sprung mass (chassis, driver, accumulator, powertrain) and unsprung mass (wheels, tires, uprights). The CG height is the mass-weighted average of the parts:

$$h = \Sigma(m_i r_i) / \Sigma m_i$$

As tested, where r_{sprung} and r_{unsprung} are the CG heights of the sprung and unsprung mass:

$$h_{\text{measured}} = (m_{\text{sprung}} r_{\text{sprung}} + m_{\text{unsprung}} r_{\text{unsprung}}) / m_{\text{total}}$$

Changing the ride height by ΔRH moves only the sprung mass. The wheels, tires and uprights stay where they are:

$$h_{\text{new}} = (m_{\text{sprung}} (r_{\text{sprung}} + \Delta RH) + m_{\text{unsprung}} r_{\text{unsprung}}) / m_{\text{total}}$$

$$h_{\text{new}} = h_{\text{measured}} + (m_{\text{sprung}} / m_{\text{total}}) \times \Delta RH$$

where $\Delta RH = RH_{\text{new}} - RH_{\text{tested}}$. Going from the tested ride height down to the design ride height, $\Delta RH = -(RH_{\text{tested}} - RH_{\text{nominal}})$, which gives the equation above. The mass ratio $m_{\text{sprung}} / m_{\text{total}}$ is the same as the weight ratio W_{sprung} / W .

An error in θ changes h by roughly $\Delta h \approx (d / 2) \times \Delta \theta / \sin^2 \theta$ ($\Delta \theta$ in radians). A 0.2° error gave roughly a 0.1 in error in h for the car measured in [2], so balance the car several times and average.

2.6 Procedure

1. Prepare the car by following the pre-test procedure.
2. Wrap the chain around one front rim and one rear rim on the same side, and close it with the maillons.
3. Attach the crane hook to the chain you just wrapped.
4. With the crane, raise the car slowly toward the expected angle from the expected CG height, watching the digital level.
Use $\theta_{\text{expected}} = \arctan((d / 2) / h_{\text{expected}})$.
5. Stand by the driver's seat as the car tilts. Put one hand on the roll hoop and one hand on the inner side of the monocoque.
These are the points where you apply force to slowly balance the car.
6. Have someone record a video of the digital level while you teeter the car back and forth. When you get it to balance at the right point, call out a word. This lets us go back through the video and read exactly what angle was measured.
7. Keep the video going and balance the car several more times to get multiple data points (about 5). Someone else can also teeter the car to get another source of data points.

Use this procedure to measure three cases:

1. Without the driver
2. With driver 1
3. With driver 2

2.7 Data and Results

Input	Value
d , width across outside tire edges at the CG (in)	
h_{expected} , expected CG height, can be from CAD (in)	
θ_{expected} (deg)	
$RH_{\text{tested}} - RH_{\text{nominal}}$, ride height raise from the dummy shocks (in)	
W_{car} , car weight without driver (lb)	
W_{unsprung} , unsprung weight (lb)	
Driver 1 weight (lb)	
Driver 2 weight (lb)	
W_{sprung} / W : without driver, driver 1, driver 2	

● Case 1: Without driver

Trial	θ (deg)	h (in)	Notes
1			
2			
3			
4			
5			
6			
7			
8			

Result	Value
Trials used	
Average h (in)	
Lowest h (in)	
Highest h (in)	
Average h at normal ride height (in)	

■ Case 2: Driver 1

Trial	θ (deg)	h (in)	Notes
1			
2			
3			
4			
5			
6			
7			
8			

Result	Value
Trials used	
Average h (in)	
Lowest h (in)	
Highest h (in)	
Average h at normal ride height (in)	

▲ Case 3: Driver 2

Trial	θ (deg)	h (in)	Notes
1			
2			
3			
4			
5			
6			
7			
8			

Result	Value
Trials used	
Average h (in)	
Lowest h (in)	
Highest h (in)	
Average h at normal ride height (in)	

3 References

1. W. F. Milliken and D. L. Milliken, Race Car Vehicle Dynamics, SAE, 1995, Section 18.2.
2. J. Hakewill, "Measuring Center-of-Gravity Height on a Formula Car," June 2007. Method credited to Claude Rouelle.
3. J. Callister and L. Sganderlla, ARG16 CG Height Measurement Testing Protocol, Cornell Racing.